A Stimuli-Responsive Hydrogel Array Fabrication Scheme for Large-Scale and Wearable Microfluidic Valving

Jiawei Tan, Haisong Lin, Shuyu Lin, Wenzhuo Yu, Jialun Zhu, Yichao Zhao, Xuanbing Cheng, Siyang Yang, Eric Tang, and Sam Emaminejad

***Abstract*— Programmable microfluidic valving enables controlled routing and compartmentalized manipulation of fluid within networks of microfluidic channels—capabilities which can be harnessed to implement an automated, massively parallelized, and diverse set of bioanalytical operations in large-scale microfluidics (lab-on-a-chip) and wearable (lab-on-the-body) applications. Stimuli-responsive hydrogels are suitable base materials to construct programmable microfluidic valving interfaces: once embedded in a microfluidic channel, their volumetric shrinkage/expansion (in response to stimulus) can be exploited to open/close microfluidic channels. However, to adapt them for the envisioned applications, critical fabrication challenges including robustness (e.g., complete channel sealing), scalability (forming arrays of valves with high yield and throughput), miniaturization of the valve actuation interface, and mechanical compatibility (flexibility for wearability) must be addressed. Here, we devise a simple and low-cost fabrication scheme to create arrays of stimuli-responsive hydrogels (e.g., thermo-responsive) and optional stimulus embodiments (e.g., microheaters) with compact footprints and within complex microfluidic networks. Within the framework of this fabrication scheme, we specifically: 1) introduced an ex situ hydrogel hydro-conditioning step to achieve full channel sealing; 2) optimized the valve performance to achieve maximal volumetric response; and 3) utilized mechanically flexible and thin device layers to ensure compatibility for wearable applications. We illustrated the scalability of the fabricated valves and their enabling microfluid management capabilities, by demonstrating fluid routing/compartmentalization within valve-gated square matrix and radial tree matrix microfluidic networks. [2020-0211]**

***Index Terms*— Large-scale microfluidics, microfluid management, microfluidic valving, stimuli-responsive hydrogels, wearable devices.**

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The authors are with the Interconnected & Integrated Bioelectronics Lab (I^2BL), Department of Electrical and Computer Engineering, University of California at Los Angeles, Los Angeles, CA 90095 USA (e-mail: tanjiawei@ucla.edu; linhaisong@ucla.edu; linsy13@ucla.edu; wenzhuoyu@ucla.edu; jialunz@ucla.edu; yzl884@ucla.edu; chengxb@ucla.edu; siyangyang@ucla.edu; erictang728@gmail.com; emaminejad@ucla.edu).

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I. Introduction

TO ADAPT stimuli-responsive hydrogels for large scale and wearable microfluidic valving, critical fabrication challenges including robustness, scalability, miniaturization of the valve actuation interface, and mechanical compatibility (flexibility for wearability) must be addressed. Previously reported hydrogel-based valve development protocols commonly relied on either 1) the ex situ construction (polymerization of the hydrogel outside the microfluidic structure) of the hydrogel features [1], which were then embedded in microfluidic channels manually, or 2) the in situ construction (polymerization of the hydrogel inside the microfluidic structure) of the hydrogel features, which were immobilized with the aid of adhesion promoters during their synthesis [2], [3]. Both approaches are unable to render full channel sealing (a prerequisite for robust valving operation). That is because of the physical mismatch of hydrogel/channel in the former approach or constraining covalent forces at the hydrogel/channel interface (limiting the hydrogel expansion) in the latter approach [2], [3]. To achieve full sealing, recent efforts utilized in situ and continuous pre-conditioning hydration steps aiming to maintain hydrogel in its expanded state [4]. However, the valves that are fabricated following the devised mechanisms are operationally incompatible for large-scale microfluidics and wearable applications: the inevitable needs for buffer exchange (to replace the hydrant solution with biofluid sample solution) and hydrant solution storage/delivery make both the design and operation of the device complicated, limiting the scalability of the device.

To overcome the aforementioned challenges, and to position hydrogel valves for large-scale microfluidics and wearable applications, we devise a simple and low-cost fabrication scheme, which allows for creating arrays of stimuli-responsive hydrogels, embedded within complex and mechanically flexible microfluidic networks with compact footprints. To illustrate the utility of our scheme, we particularly leveraged the poly(N-isopropylacrylamide) (PNIPAM) hydrogel, which can be thermally actuated via miniaturized heating elements (for instance, by heating elements on a circuit board interfacing the microfluidic module, or by microheaters directly integrated with the microfluidic module). We illustrated the scalability of the fabricated valves and their enabling microfluid management capabilities, by demonstrating fluid routing/compartmentalization within various microfluidic configurations.

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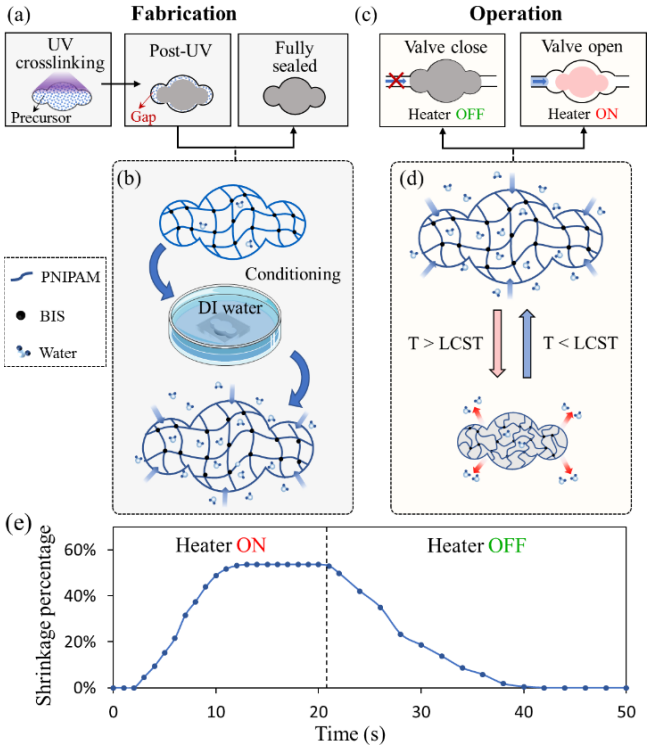


Fig. 1. (a) Fabrication of a stimuli-responsive hydrogel-based valve. A one-step hydrogel hydro-conditioning step (in the deionized, DI, water) is performed after UV crosslinking and prior to the incorporation of the hydrogel in the channel to ensure full channel sealing. (b) Illustration of the expansion of the hydrogel due to the infusion of water molecules. (c) Schematic operation of the PNIPAM hydrogel valve: activation of the heater results in hydrogel shrinkage, opening of the channel, and permitting the fluid flow. (d) Illustration of the PNIPAM hydrogel shrinkage/expansion upon adjusting the temperature above/below the hydrogel's LCST. (e) Characterization of a representative PNIPAM hydrogel's temporal volumetric response to the activation and deactivation of the heater.

II. Results and Discussion

A. Stimuli-Responsive Hydrogel Valve Array Fabrication

Our fabrication scheme consists of the following main steps: 1) laser patterning polyethylene terephthalate (PET)/double-sided tape substrates to define 2D valve and channel features in designated "valve" layer and "microfluidic channel layers" (as two separate layers), respectively; 2) polymerizing hydrogel in situ (with optimized crosslinker concentration), via exposing the valve layer to ultraviolet (UV) light (Fig. 1a); 3) hydro-conditioning the valve layer to achieve hydrogel expansion (due to infusion of water molecules), and render full channel sealing (Fig. 1b); and 4) aligning and integrating the open-face valve and microfluidic channel layers to vertically connect the 2D fluidic pathways between the two layers, realizing a complete valve-gated microfluidic channel. We particularly constructed our valve, using the thermo-responsive PNIPAM hydrogel, which reversibly shrinks and expands in response to temperature increase/decrease around its lower critical solution temperature (LCST) [5] as illustrated in Fig. 1c, d.

The reversible thermo-responsive property of the PNIPAM hydrogel originates from the temperature-tunable interactions between water molecules and the hydrophilic amide group/hydrophobic propyl group within its polymer structure [6]. As shown in Fig. 1e, in the context of temperature change induced by a local heater, the PNIPAM-based hydrogel shrunk 55% when activating the heater and restored to its initial size when deactivating the heater.

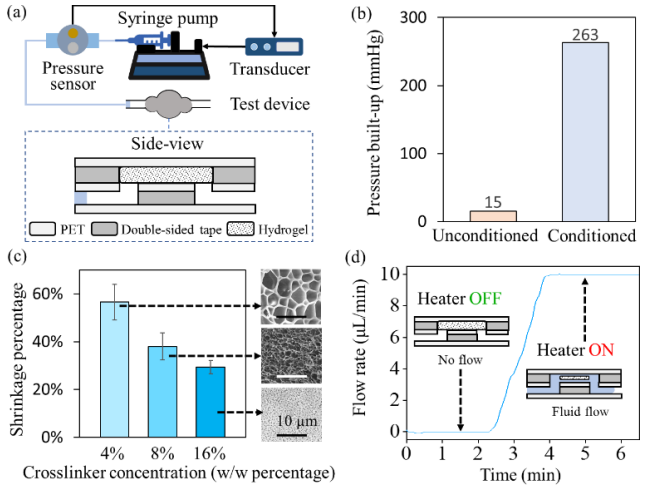


Fig. 2. (a) Pressure sensor-coupled syringe pump system setup used for the characterization of the test devices. (b) Back pressure characterization of the hydrogel valves, fabricated with and without the conditioning step. (c) Shrinkage percentage of hydrogels, fabricated with different crosslinker concentrations ($n = 3$, error bars indicate standard error). Insets show the SEM images of the corresponding hydrogels. The lower the crosslinker concentration, the larger the resultant pores in the hydrogel structure, leading to the higher rate of water molecule diffusion and the larger hydrogel volumetric response to temperature changes. (d) Characterization of flow rate through a hydrogel valve-gated microfluidic channel. Upon turning the heater on, and subsequent hydrogel shrinkage, the fluid will flow in the channel and pass through the valve chamber through vertically aligned VIAs (vertical interconnect accesses).

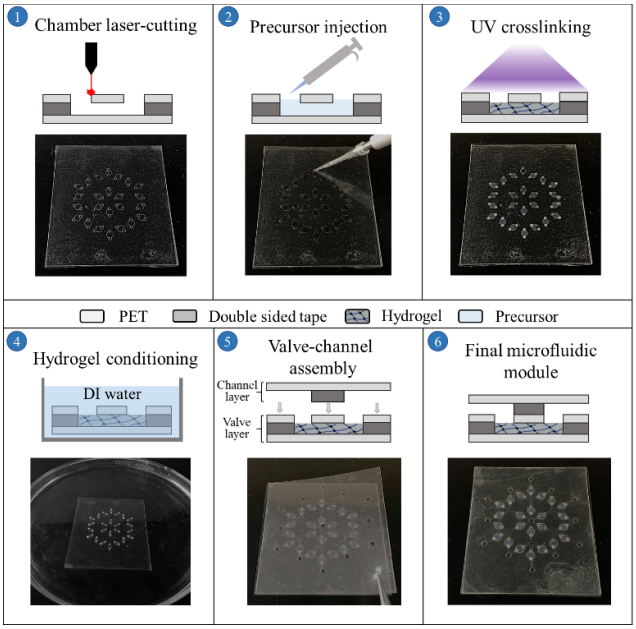


Fig. 3. Fabrication of a hydrogel valve-gated microfluidic network. The hydrogel valve array and microfluidic channel features are patterned by laser-cutting (~10 μm precision) the valve and channel layers. Next, the PNIPAM precursor solution is injected into the designated valve chambers (on the valve layer) through the vertical openings, and then polymerized via UV crosslinking. The hydrogel array is conditioned via immersion of the valve layer in DI water for 12 hours. Then, the open-face valve and channel layers are vertically aligned and assembled. PET thickness: 100 μm, PET flexural modulus: 3380MPa, double-sided tape thickness: 170 μm. The size of the valve is 5 mm (axially).

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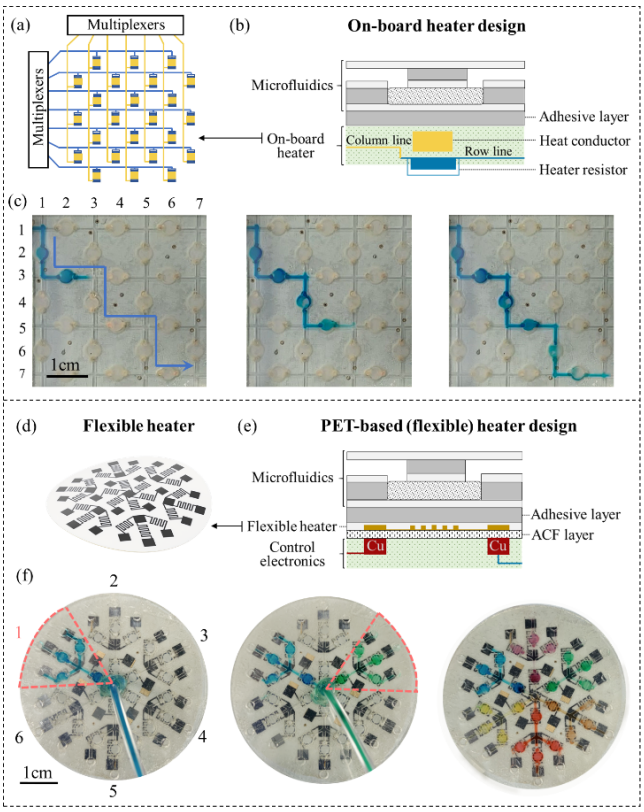


Fig. 4. (a-c) Fluid routing within a square matrix microfluidic network. (a) Schematic of the addressable on-board heater array. The multiplexers facilitate the electrical connections of the selected valve's heater terminals. (b) Side-view schematic of the valve-gated microfluidic channel and the heater interface. (c) Sequential optical images illustrate fluid routing through a zigzag path. (d-f) Fluid routing and compartmentalization within a radial tree matrix microfluidic network. (d) Schematic of the addressable flexible (PET-based) heater. (e) Side-view schematic of the valve-gated and heater-integrated microfluidic module. The heaters are electrically connected to control electronics via an incorporated anisotropic conductive film (ACF) layer. (f) Sequential optical images illustrate fluid routing and compartmentalization.

We evaluated the valving performance, with regards to channel sealing (when valve closed), by using the pressure characterization setup shown in Fig. 2a. As illustrated in Fig. 2b, the devised ex situ hydro-conditioning step resulted in increasing the valve's back pressure (maximum pressure that the valve can hold without any leakage) by more than 15 times. To achieve the highest degree of valving opening, the shrinkage percentage of the hydrogel was optimized by adjusting the crosslinker concentration (N, N′ -methylenebisacrylamide, BIS). As shown in Fig. 2c, the hydrogel with 4% crosslinker exhibited the highest shrinkage (∼55%), as compared to 8% and 16% cases. Fig. 2c's insets show the corresponding scanning electron microscopy (SEM) images of the hydrogels with different crosslinker concentrations. The lower crosslinker concentrations (<4%) yielded more fragile hydrogels, which were unreliable for valving. The valve-gated microfluidic channel flow characterization results, shown in Fig. 2d, validated the achievement of the desired binary valve operation: fluid flow is completely blocked with no leakage (flow rate: 0 μL/min) when the valve is deactivated, and fluid flow is permitted when the valve is activated (flow rate: 10 μL/min).

The scalability of our fabrication procedure allows for simply extending the developed and optimized PNIPAM-based valve into an array format with different arrangements. The detailed steps of our fabrication scheme are shown in Fig. 3. In particular, our scheme's capability in rendering valve/channel features with micrometer-spatial resolution within disposable, thin, and mechanically flexible microfluidic modules, makes it suitable for constructing both large scale microfluidics and wearable platforms.

B. Fluid Routing and Compartmentalization Within Valve-Gated Complex Microfluidic Networks

The PNIPAM hydrogel valve arrays can be thermally actuated via addressable miniaturized heating elements. Toward positioning our valve arrays for large-scale microfluidic and wearable applications, we particularly realized the corresponding addressable actuation interfaces on: 1) a multi-layered printed circuit board, featuring highly dense heating elements, connected to control programmable circuitry (Fig. 4a-c), and 2) a flexible substrate (PET), directly integrated within the microfluidic module, forming a fully flexible device (Fig. 4e-f). Following these implementations, we illustrated biofluid routing and compartmentalization within a valve-gated square matrix and a radial tree matrix microfluidic network.

III. Conclusion

Here, we devised a simple and low-cost fabrication scheme to create stimuli-responsive hydrogel valves, addressing fabrication challenges, such as robustness and scalability. To demonstrate the utility of our scheme, we particularly selected/optimized a thermo-responsive hydrogel. To illustrate its potential for fabricating wearable and large-scale microfluidics platforms, we extended our scheme to fabricate thermo-responsive hydrogel valve arrays—coupled with actuation interfaces (on-board/flexible microheaters)—within complex microfluidic networks. The demonstrated biofluid management capabilities (e.g., fluid routing and compartmentalization) can be adapted to implement automated, massively parallelized, and diverse bioanalytical operations in large-scale microfluidics (lab-on-a-chip) and wearable (lab-on-the-body) applications.

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